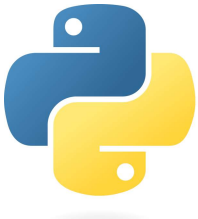
Python Wiki - Toolbox and Library Installation



TLDR for most 2026 Spring Windows students

1. Install **Python 3.12.10** (https://www.python.org/downloads/release/python-31210/?wrap=1) ↓ (https://www.python.org/downloads/release/python-31210/download?download_frd=1) using the Windows installer (64-bit). Tick **Add python.exe to PATH** during installation.
2. Install **Visual Studio Code** (https://code.visualstudio.com/?wrap=1) and, if required, **Microsoft Visual C++ 14.0 or higher** (https://visualstudio.microsoft.com/visual-cpp-build-tools)
3. Open your subject folder in VS Code, then create a virtual environment:

```
py -3.12 -m venv .venv
```

4. **Activate the virtual environment in VS Code:**
   - Press `Ctrl + Shift + P` to open the Command Palette.
   - Type and select **Python: Select Interpreter**.
   - Choose the option that points to your `.venv` folder.
   - **Crucial:** Close your current terminal by clicking the **Trash Can icon** in the top-right of the terminal panel (VS Code cannot update an already open terminal).
   - Open a **New Terminal** (`Ctrl + Shift + ``).
   - **Verify:** Make sure you see `(.venv)` at the very beginning of your terminal prompt before moving to the next step.
5. Install the subject support package:

```
python -m pip install ir-support-full
```

This installs IR Support, Peter Corke's Robotics Toolbox dependencies, Swift, and the extra robot and object model packs.

6. On Windows, apply the current subject workaround patches:

```
python -m ir_support.doctor --patch
```

The doctor command checks first and only changes installed package files when known Robotics Toolbox, Swift, PyPlot, or Machine Vision fixes are needed.

Estimated download: allow ~500 MB for Python packages and model packs, plus several GB if Windows needs Microsoft C++ Build Tools.

For details, or for macOS/Linux notes, see the **Python Wiki - Bug Fixes and FAQ (https://canvas.uts.edu.au/courses/39957/pages/python-wiki-bug-fixes-and-faq)** page.

Detailed Instructions

Install Python in your chosen Operating System

Windows:

The recommended version for new 2026 installs is **Python 3.12.10** (https://www.python.org/downloads/release/python-31210/?wrap=1) (https://www.python.org/downloads/release/python-31210/download?download_frd=1), installed via the "Windows installer (64-bit)". Python 3.12.10 is the last Python 3.12 release with normal Windows installers; later Python 3.12 security releases may exist, but may not provide Windows installers. The current IR Support release has been checked with Python 3.10, 3.11, and 3.12.

Do not use Python 3.13 for this subject. The Robotics Toolbox and RVC3 package stack used in 41013 currently targets Python versions up to 3.12.

If you haven't already installed VS Code, it is recommended as your IDE — although you may use any Python IDE you prefer.

If Python is already installed via VS Code, make sure to select the recommended Python interpreter:

- Press `Ctrl+Shift+P`
- Type "**Python: Select Interpreter**"
- Choose the interpreter for your subject environment. For new installs, this should usually be **Python 3.12**.

Then, in the VS Code terminal (`Ctrl+~`), confirm the selected version:

```
python -V
```

Ensure it shows **Python 3.12** for a new install, or another supported version from **Python 3.10 to 3.12**, and that the path points to the correct install.

If it still does not work (e.g., it shows the wrong version or fails to find Python), but you are sure Python is installed, check the following:

- If you get an error like "*Python was not found; run without arguments to install from the Microsoft Store, or disable this shortcut from Settings > Apps > Advanced app settings > App execution aliases*", you probably didn't check the box "Add python.exe to PATH" when you installed. You can follow the instructions to disable this shortcut. If you cannot find this in Settings, open the **Start menu** and type **"Manage App Execution Aliases"** to go there directly and toggle off the Python shortcut to the Microsoft Store. Then manually add the Python main directory and the Scripts directory to your user environment variables. Close and reopen VS Code and try again.
- Alternatively, check via the Windows "Apps & Features" menu, then try this in the terminal — replacing `USER_NAME` with your Windows username:

```
C:\Users\USER_NAME\AppData\Local\Programs\Python\Python312\python.exe -V
```

If that returns the correct version, you can configure the terminal to always use it by editing your VS Code settings:

1. Press `Ctrl+Shift+P` and select "**Preferences: Open Settings (JSON)**"
2. Add the following line (replace `USER_NAME` with your actual username):

```
"terminal.integrated.env.windows": {"PATH": "C:\\Users\\USER_NAME\\AppData\\Local\\Programs\\Python\\Python312;${env:PATH}"}
```

Save the file and restart VS Code. Open a new terminal and run `python -V`; it should now return your selected supported Python version, normally **Python 3.12** for a new install.

Microsoft Visual C++ 14.0 or higher

Windows users must install Microsoft Visual C++ 14.0 or higher to build certain Python packages (like the Robotics Toolbox).

To check if it's already installed:

- Open the **Start menu** and search for **Developer Command Prompt for VS 2022**
- Open it and type: `cl`
- If installed, you'll see output like:
  `Microsoft (R) C/C++ Optimizing Compiler Version 19.44...`
  (Any version **14 or higher** is fine — 19.44 is the current version as of mid-2025)

If it's not installed, follow these steps:

- Visit: **https://visualstudio.microsoft.com/visual-cpp-build-tools/ (https://visualstudio.microsoft.com/visual-cpp-build-tools/)**
- Download and launch the installer.
- In the installer, make sure you check:
  - **"Desktop development with C++"**
  - You can uncheck most Optional components (**see here (https://canvas.uts.edu.au/courses/39957/files/11910574?wrap=1) (https://canvas.uts.edu.au/courses/39957/files/11910574/download?download_frd=1)**), but it will be several GB in size because you do need:

✓ MSVC v143 - VS 2022 C++ x64/x86 build tools

✓ Windows 10 (*or 11*) SDK (10.0.19041.0) or higher

✓ C++ CMake tools for Windows

- Click **Install** and let it finish.

Installing or Upgrading pip

For a normal Python install, `pip` should be installed automatically. To check, type:

```
python -m pip -V
```

If it's not installed, or you want to upgrade to the latest version, follow these steps:

1. **Install or upgrade pip**

```
python -m ensurepip --upgrade
```

2. **Upgrade pip to the latest version**

```
python -m pip install --upgrade pip
```

3. **Check that pip is working correctly**

```
python -m pip -V
```

You should see output that points inside your selected Python installation or virtual environment, confirming that `pip` is linked to the correct Python version.

If you still see the wrong version (e.g., pointing to a different Python folder), it may be due to:

- An older `pip` still in your system PATH
- VS Code using `pip` from another environment (e.g., PlatformIO)
- A problem with your Python installation

If you're unsure, just continue using `python -m pip` instead of `pip`. This ensures you're using the correct version, or you can create a virtual environment as explained below. If you're still stuck, ask our tutors in class or tag **@Tutors** in Teams.

Linux:

Use Python 3.10, 3.11, or 3.12 on Linux. Python 3.12 is recommended for new installs where available. Python 3.13 is not recommended for this subject.

```
sudo apt-get install python3
```

Mac:

Python 3 is included with Xcode, Apple's developer toolkit. You can install Xcode via the App Store (note: ~14GB).

Alternatively, install Python via Homebrew. Follow this guide: [https://docs.python-guide.org/starting/install3/osx (https://docs.python-guide.org/starting/install3/osx?wrap=1) ↓ (https://docs.python-guide.org/starting/install3/osx/download?download_frd=1)]

Virtual Environments in Python (optional, but recommended)

To avoid conflicts with other projects, you can use a virtual environment (venv). It creates an isolated space with its own packages and dependencies. *See: **Python Tips and Tricks (https://canvas.uts.edu.au/courses/39957/pages/python-wiki-tips-and-tricks?module_item_id=2573633)***

Guide: [https://packaging.python.org/en/latest/guides/installing-using-pip-and-virtual-environments/#creating-a-virtual-environment (https://packaging.python.org/en/latest/guides/installing-using-pip-and-virtual-environments/?wrap=1#creating-a-virtual-environment) ↓ (https://packaging.python.org/en/latest/guides/installing-using-pip-and-virtual-environments/download?download_frd=1#creating-a-virtual-environment)]

Windows and VS Code virtual environment details (with auto-activation)

1. **Open your project folder in VS Code**
   (e.g., C:\Robotics_41013_Python). Note that VS Code only auto-detects virtual environments in folders you open.
2. **Create a virtual environment inside that folder**

```
py -3.12 -m venv .venv
```

3. **Select the virtual environment interpreter**
   - Press `Ctrl+Shift+P`
   - Type **"Python: Select Interpreter"**
   - Choose the interpreter that includes `.venv` and shows your supported Python version, normally **Python 3.12** for a new install.
4. **Open a new terminal (`Ctrl+~`)**
   VS Code should automatically activate the virtual environment or ask you if you want to use it. You may see the terminal prompt change to:

```
(.venv) C:\Robotics_41013_Python>>
```

5. **Install the subject support package using pip:**

```
python -m pip install ir-support-full
```

This installs the UTS IR Support Package, the required robotics toolbox dependencies, and the extra robot and object model packs. From now on, every time you open this folder in VS Code and

launch a terminal, the correct Python version and pip will be used automatically.

On Windows, after installing the package, run the doctor command to apply the known upstream toolbox workarounds if they are needed:

```
python -m ir_support.doctor --patch
```

Packages

Peter Corke's Robotics Toolbox

First, ensure that Python 3 and pip (or pip3 for Linux/Mac), and MS Visual C++ 14.0 or higher on Windows, are installed.

The UTS support package below installs the required combined Robotics, Vision & Control **rvc3python (https://github.com/petercorke/RVC3-python)** toolbox for this subject. You normally do not need to install Peter Corke's Robotics Toolbox separately if you install `ir-support` or `ir-support-full`.

If you need to repair or update the subject environment, update the subject support package family rather than force-reinstalling Peter Corke's packages directly:

```
python -m pip install --upgrade --no-cache-dir ir-support-full
```

A direct force-reinstall of `rvc3python` may let pip upgrade dependencies, such as `matplotlib`, beyond the versions currently tested with `ir-support`. Only install or update `rvc3python` separately if a tutor or the subject team specifically asks you to do so.

The combined install includes the **roboticstoolbox-python (https://github.com/petercorke/robotics-toolbox-python)** toolbox, which is the main toolbox that we use in this subject. You can check your version with:

```
python -c "import roboticstoolbox as rtb; print(rtb.__version__)"
```

The UTS Industrial Robotics (IR) Support Package

UTS wrote this package family, starting with **ir-support (https://pypi.org/project/ir-support)**, which contains helper classes/functions and the models required for the labs. Most students should install the full package with:

```
python -m pip install ir-support-full
```

The current core release is **ir-support 1.4.3**. It supports Python 3.10 to 3.12 and installs the Robotics Toolbox, RVC3, Swift, and other required dependencies for the subject.

If you only need the smaller core package, without the extra robots and parts, install:

```
python -m pip install ir-support
```

If you only want one optional pack, install it separately:

```
python -m pip install ir-support-extra-robots
python -m pip install ir-support-extra-parts
```

To update during the session, if we make changes that you need, run:

```
python -m pip install --upgrade --no-cache-dir ir-support
```

For the optional full model set, run:

```
python -m pip install --upgrade --no-cache-dir ir-support-full
```

The current package family is **ir-support 1.4.3** with **ir-support-extra-robots 0.2.1**, **ir-support-extra-parts 0.2.1**, and **ir-support-full 0.2.4** (July 2026).

Still Having Trouble?

Check out our **Python Wiki - Bug Fixes and FAQ (https://canvas.uts.edu.au/courses/39957/pages/python-wiki-bug-fixes-and-faq)** page.

Now what?

- To unlock **Quiz 1**, **complete a basic training module (https://canvas.uts.edu.au/courses/39957/assignments/267463)** and upload your completion certificate (PDF or image) to the linked Canvas assignment.
- Start with the lab starter, **Lab 1 Starter (https://canvas.uts.edu.au/courses/39957/pages/lab-1-starter?module_item_id=2573647)** and/or the **first week 1 module (https://canvas.uts.edu.au/courses/39957/modules/316576)**, and then proceed to **Lab 1 Exercise (https://canvas.uts.edu.au/courses/39957/pages/lab-1-exercises)**.